UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Intelligent Data Science and
Artificial Intelligence Research Center

Universitat Politècnica de Catalunya

Bachelor Degree on Artificial Intelligence

PROCESSOS D'ANÀLISI INTEL·LIGENT DE DADES

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Practical TeamWork: Building an Intelligent Decision Support System (PW2, PW3, PW4)

The main goal of this work is to develop a *prototype* of an Intelligent Decision Support System in a concrete domain of application. The main steps that students must undertake are listed below.

Procedure

1. *Form a working team*. Each working team should be composed of 4 to 6 members.
2. Select which is the *domain of application* or *problem* where the IDSS will be applied, and *identify the main decisions* to which the IDSS will give support. There are some possibilities like:
    a. Select one domain or scenario from the PW1 of one of the team members
    b. Select one domain from an open data repository (see list of websites available at the open data documents in the course website), and identify a decision to be assessed using these data, so defining the associated *IDSS problem*.
    c. Select a new domain of application/problem which is interesting for all the members of the team. Ensure that it is *an IDSS problem*.

    For options a and c be carefull to think about a problem where data is available
3. Use the AI-Council Canvas to design the scope and approach of your IDSS (find the CANVAS template in the course web)
4. Model and design the decisions you want to support with your system. It is interesting that the IDSS you propose contains **at least 1 Data-driven model and 1 knowledge-based component**
5. Identify the type of users of your system and the functionalities required for each of them. *Example: Admin configurates the system, control users registration; Politician makes queries and receives result; Administrative Staff introduces data*
6. Derivate the architecture of the IDSS as a particular case of the general architecture shown in class.
7. Describe the *database/s, model-based techniques* and/or *expert knowledge* available for your problem
8. Determine the software tools to be used (R, Weka, RapidMiner, etc.) and the

programming languages to be used (Python, Java, C++, etc.)

9. Design and implement the data preprocessing process to prepare the data for the training: provide a global flow diagram for the entire *data pre-processing process*
10. Built the different *data-driven models* (descriptive, discriminant, predictive, associative) from your training data.
11. Built the knowledge components of your IDSS *(ex, expert-based models* or *agent-based simulation models* or *qualitative reasoning models* or *model-based reasoning techniques* (*model-driven IDSS techniques*, *CBR reasoning engine, a fuzzy reasoner, or an ontology…)*
12. Test and validate the correctness of each data-driven or knowledge-driven model induced from the training step.
13. Design the *user interface* of your IDSS and connect all modules according to the architecture designed in step 6, including *the induced or obtained models* into your system. Take care to introduce in the IDSS **the result** of the data-driven modelling process*. Example: If you train a decision tree model, the result is a set of machine-readable rules, that can be evaluated in front of new data at production time.*
14. Design 3 use cases or more that show the applicability of your IDSS to give support to the different decision-making processes of your application domain. This means to imagine 3 different real situations where your IDSS will work and describe who queries the system and how and which will be the outputs.
15. Prove the performance of your system in the designed scenarios and prepare a demo.
16. *Evaluate the global Intelligent Decision Support System prototype* constructed.
    a. Describe the design of the testing to evaluate the technical aspects
        - Evaluate how the models perform
        - Evaluate the integration of components

        and all other aspects you consider relevant
    b. Include in the evaluation an additional assessment where you evaluate the appropriateness of your IDSS in regards of the following topics:
        - Ethics in AI
        - Explainability
        - Gender perspective
        - Digital gap

        For each of these topics explicitly describe which specificities of your design guarantees the fulfillment of it, strengths and drawbacks of your system in this regards and risks.
17. Describe a plan of implantation of your IDSS in a real setting
    a. In which part of the process the IDSS will be inserted?
    b. Who will access the system and with which privileges?
    c. How the results are used by the different users?
    d. Steps of adoption. Is the system put in production completely in a certain date? Are some transformation of data infrastructures required before? Do the users require previous training? Etc etc

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Deliverables

The delivery dates of each delivery are shown in the schedule of the course

PW2-D0: Choose a working Team and communicate
Once the groups consolidated and approved by the lecturer, the leader of the working team must send an email to the coordinator of the course in CC with the laboratory instructor including CC with mails of **ALL members** of the working team.

The **subject of the email** must be:

"PAID GIAdegree WT <number>"
(ex. PAID GIAdegree WT 3)
The number is the working team number previously assigned to your group by the instructor.

This will be used as the basic communication reference between the instructor and the working team. From that point on, be sure that **all** questions mailed to the instructor uses this complete mailing list.

PW2-D1: Describe your working topic
Present a one page report with the following information:

1. Name of all group components by alphabetical order (sort by Family name)
2. Brief description of the goals of the proposed IDSS
3. Which decisions will be supported
4. Which data will be used
    a. Basic structure of every data matrix involved:

Data source (url or company, etc)	**Nr of records (2000-5000 is enough)** if you are working with countries in the world or other situations with less, this might be reconsidered, check with the instructor)	**Nr of variables**	**Nr of numerical variables (minimum 7)**	**Number of binary variables (minimum 2)**	**Number of other qualitative variables (mínimum 5)**	**% of missing data in the dataset**

b. For each dataset involved in the previous table include: One paragraph explaining what data are about
Also, fill in the following template for each dataset involved in your system. It is possible to enlarge your data base with additional variables comming from other sources if you like, but do not invest too much time on that

Data Source (url or company, etc)	Name of Variables	Type (Num, Qual, et)	Granularity/frequency (t real, online, ...)	Nature (Open, private, ...)	% of missing data of the variable	How to obtain the data in training step (basic download, on-line, query SQL, etc)	How to obtain the data in production step (basic download, on-line, query SQL, etc)

5. Global idea about the knowledge components you think will be used in the system and information sources you intend to use with this components (literature, experts, etc)

PW2-D2: AI-Council Canvas of your project. Find the template in the course website and describe the use case you address in your project.

PW2-D3: Enlarge the PW2-D1 with the AI-Council Canvas and the chapters corresponding to

1. IDSS definition,
2. Decision model (point 4 of the general work description above)
3. Brief description of types of users
4. IDSS architecture.
5. Description of information sources (data/knowledge)
6. Description of software tools selected for the work
7. Design of entire data preprocessing process
8. Results of the entire preprocessing
    a. Metadata
    b. Preprocessed data matrix

PW2-D4: Oral presentation (delayed to the PW3-D2 date), (15 min presentation + discussion with instructors)

PW3-D1: Enlarge the PW2-D3 with the chapters corresponding to

1. Data-driven models included in your system. For each of them
    a. Configuration of the training parameters
    b. Characteristics of the training data
    c. Results of the model validation

   d. Description of the trained model and how to integrate in the entire system
2. Knowledge representation details: Justify the knowledge formalisms used to inject knowledge into the system and the added value of these knowledge. For each knowledge component of your system describe:
   a. Formalism (ontology, Knowledge base, etc)
   b. Main structure
   c. Source of contents (experts, literature, data-driven models, etc)
   d. Role in the system
   e. Form of the output, results and model validation
   f. Describe how to integrate into the general architecture

3. Reasoning components. Description of the nature and role of the reasoning components if any. Also describe the integration into the general architecture

PW3-D2: Oral presentation (15 min presentation + discussion with instructors)

PW4-D1: Enlarge PW3-D1 with the front-End, IDSS validation and prove of concept

1. I/O
   a. Describe the types of users and permissions and roles
   b. Describe the inputs of the system (data, etc)
   c. Design and implement of the front-end screens
   d. Describe the ouputs of the system for each type of user and functionality
   e. Describe the mecanisms of explainability of the system
   f. Describe the interactions front-end/back-end
   g. Integrate all system components
   h. Implementation and validation of all interactions (provide a validation protocol properly designed before the prove of concept)
      i. Ethics in AI assessment
      ii. Gender perspective assessment
      iii. Digital gap assessment
2. Describe the 3 use cases designed. For each of them, report the performance of the system obtained.
3. Prove of concept and demo preparation
4. Describe the plan of the launching of the system

PW4-D2: Oral presentation of the second part of the work (15 min presentation + discussion with instructors)

At every delivery a **ZIP file** labelled as "**PW234-PAID-YearCourse-T#NumberOfTeamwork-TitleOfWork**", should be uploaded **through** "**Racó de la FIB**" in due time with the content bellow. At every delivery include the corresponding parts and growth the materials along the course.
(only the head of the working team submits to the raco)

1. A folder named "**Documentation**" with:
    a. The report containing PW2-D1 to PW4.D1 plus:
    b. A final section on Future work and improvements
    c. Gantt diagram with tasks planning of the project development
    d. Tasks assignment and responsibilities among teamwork members
    e. Time sheet, with hours devoted by each teamwork member to each task, andglobal accounting by each teamwork member
2. A folder named "**Data**" with the files with the original raw dataset/s or database/s and preprocessed ones used for *data-driven models*, both for training and for testing.
3. A folder namet "**KnowledgeSources**" with the knowledge sources of the system (ontologies, Knowledge bases, etc)
4. A folder named "**Models**" with the *data-driven models* or/and *model-driven techniques* used which can be visualized, read or interpreted in a nice way.
5. A folder named "**Reasoning engines**" if any
6. A folder named "**Source**" containing the source code of the scripts/macros/ functions used in the different software tools/languages used at training step.
7. A folder named **"IDSS"** containing the source code and the exe (if any) of the final system
8. A folder named "**Demo**" with a 3-5 minutes video with the demo.
9. A folder named "**Presentation**" containing the slides (up to 15) of the public presentationof the work:
    a. One title slide with the complete title of your work, your group number and names of all members of the group and date.
    b. Several slides summarizing the report delivered. Be sure to include a slide with the kind of actors interacting with your IDSS.
    c. Be sure to include a slide with the AI Council Canvas of your IDSS.
    d. Be sure to include a slide with the kind of actors interacting with your IDSS.
    e. Be sure to include a slide with the architecture of your IDSS.
    f. Be sure to include a slide describing the 3 use cases tested.
    g. Deails on the different components of the system

10. A **README.txt file** specifying the structure and contents of the ZIP file.

Students must deliver the ZIP file on **PW4-D1** through the Racó
The public communication of the contents and general discussion with all the students in the course will be done on **PW2-D4, PW3-D2, PW2-D2**

Several items will be considered for the qualification of this work:

- The quality of the methodologyand work done.
- The documentation delivered,
- The quality of the oral exposition (both presentation and content assessed, as well as the ability to answer questions),
- The planning, coordination andmanagement of the team, and
- The individual evaluation of each student, including her/his integration level within the team group.